Unsupervised Learning : Clustering

IS 2210

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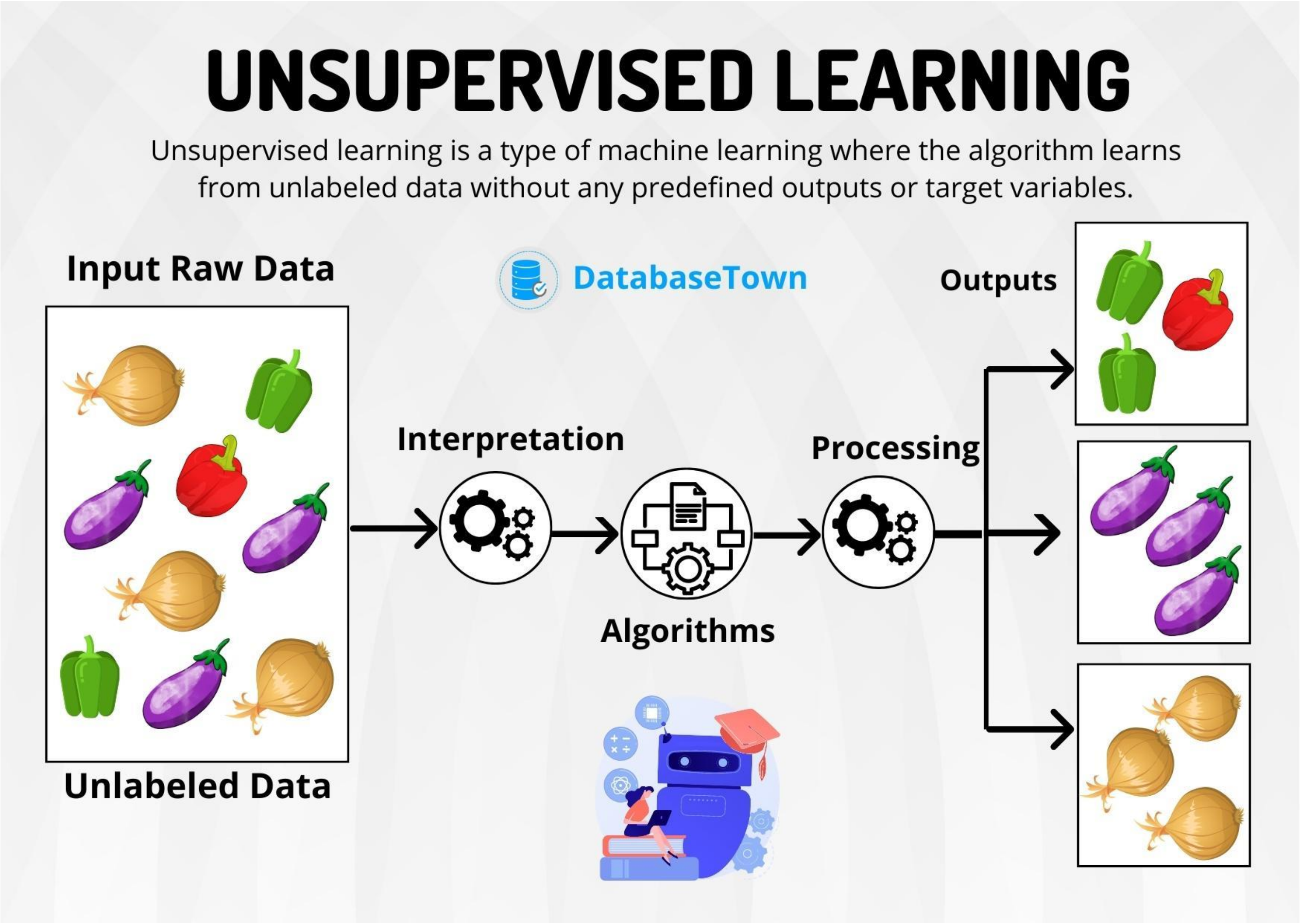
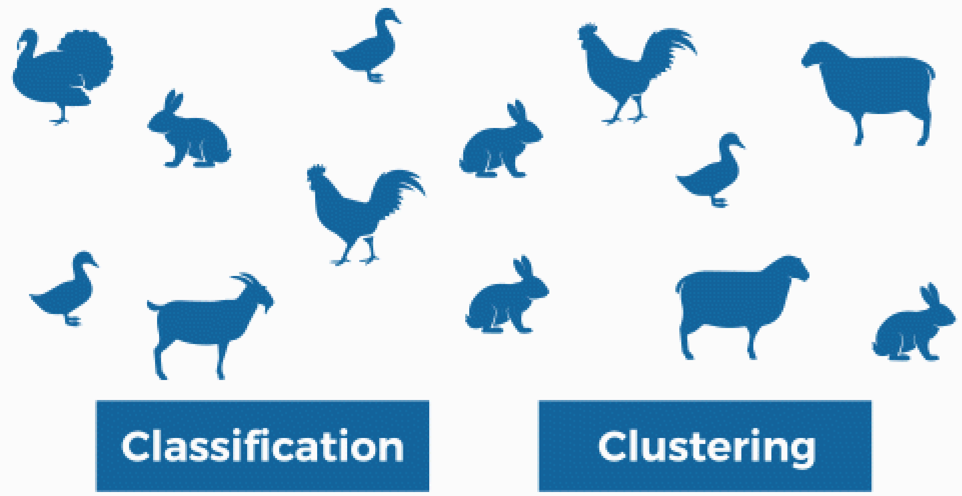
Learning outcomes

- Explain the difference between supervised and unsupervised learning.
- Define clustering and describe its real-world applications.
- Explain and implement K-means clustering.
- Interpret clustering results using evaluation metrics
- Describe hierarchical clustering and interpret a dendrogram.


UNSUPERVISED LEARNING
Unsupervised learning is a type of machine learning where the algorithm learns from unlabeled data without any predefined outputs or target variables.
Input Raw Data
DatabaseTown
Outputs
Interpretation
Processing
Algorithms
Unlabeled Data

Unsupervised Learning? - Recap

- Algorithm learns without having desired output for a given input.
- It just attempts to find out the structure within the data set (identify similarities and divisions among data).
- With unsupervised learning, a model has no established guidelines for desired outputs or relationships. Instead, the goal is to explore the data and, in doing so, discover patterns, trends, and relationships.
- The main advantage of unsupervised learning is that the training data does not need to be labeled.
- All the data is considered input, and the output would typically be information about the inherent structure of the data


Classification
Clustering

Clustering

- Clustering is an unsupervised learning technique in machine learning that involves grouping similar data points into clusters or categories without prior knowledge of class labels.
- It seeks to uncover hidden patterns and structures within datasets by analyzing data similarities and differences.

Real world applications

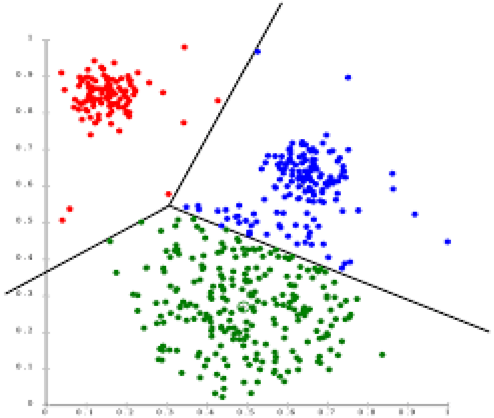
- Grouping the content of a website or product in retail business
- Segmenting customers or users in different groups om the basics of their metadata and behavioral characteristics (Customer segmentation)
- Segmenting communities in ecology.
- Finding clusters of similar genes.
- Image segmentation.
- Identify outliers or unusual patterns in data, which is critical for fraud detection and network security (Anomaly Detection)

Types of Clustering methods

- **Centroid-based Clustering**
- **Hierarchical Clustering**
- Density-Based Clustering
- Distribution-Based Clustering

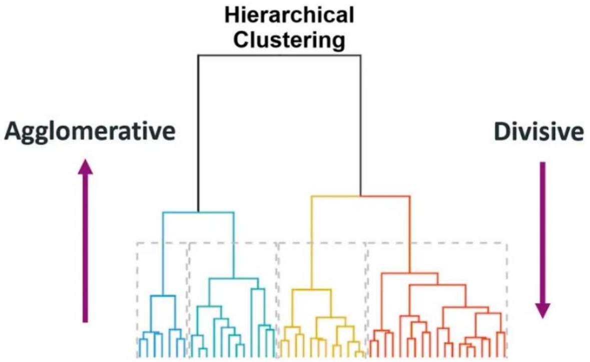
Centroid-based Clustering

- Centroid-based Clustering aims to partition a dataset into groups or clusters based on the proximity of data points to the centroid of each cluster.
- The centroid of a cluster is the arithmetic mean of all the data points in that cluster.
- Centroid-based clustering has several variations :
  - K-Means Clustering
  - K-Medoids Clustering
  - Fuzzy c-Means Clustering



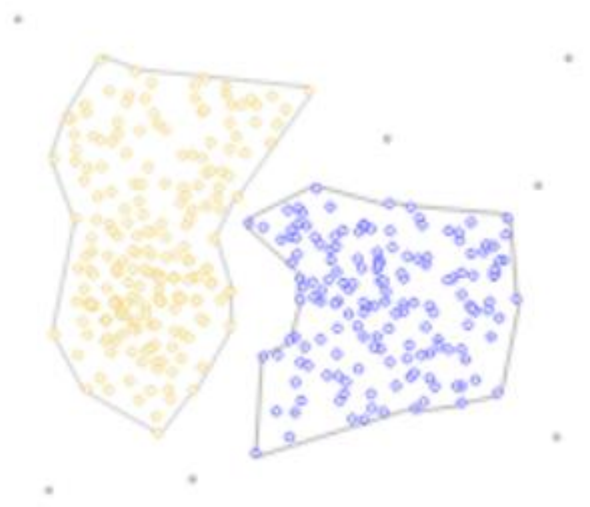
Hierarchical Clustering

- It creates a hierarchy of clusters by merging or splitting them based on similarity measures.
- It creates a dendrogram a tree-like structure that reflects relationships at various granularity levels .
- It uses a bottom-up or top-down approach to construct a hierarchical data clustering schema.
- It does not require specifying cluster numbers in advance.
- There are mainly two types of hierarchical clustering:
  - Agglomerative hierarchical clustering
  - Divisive Hierarchical clustering



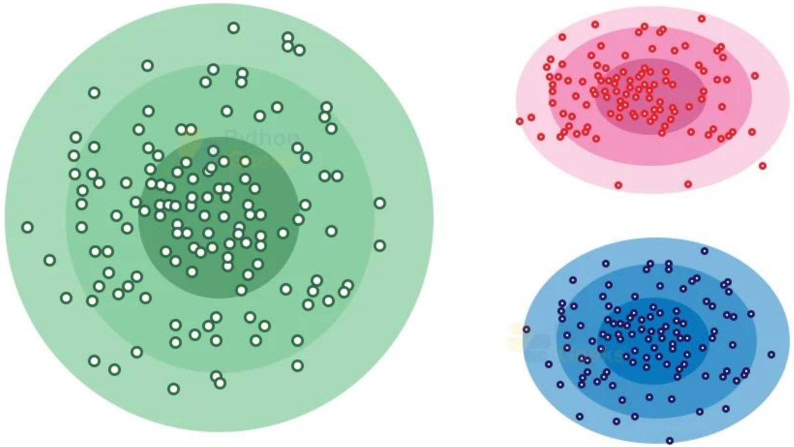
Density-Based Clustering

- Density-Based Clustering groups data points based on areas of high density separated by areas of low density.
- Basically, the algorithm finds the places that are dense with data points and calls those clusters.
- Most common density-based clustering algorithms :
    - DBSCAN Clustering
    - OPTICS Clustering



Distribution-Based Clustering

- All of the data points are considered parts of a cluster based on the probability that they belong to a given cluster.
- It assumes data is generated from a mixture of probability distributions, such as Gaussian distributions and assigns points to clusters based on statistical likelihood.
- Example :
    - Gaussian Mixture Model (GMM)



K-Means Clustering

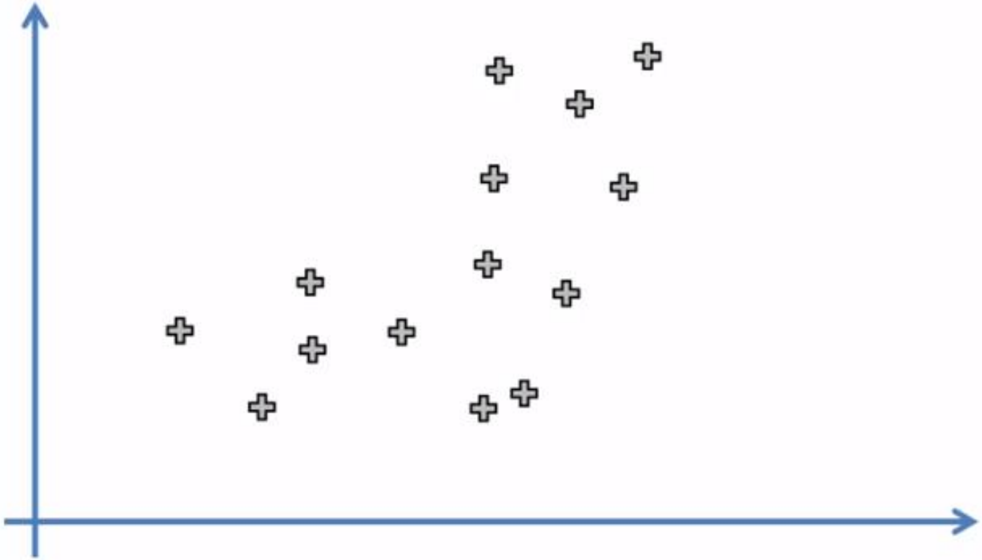
- An iterative algorithm that splits a dataset into non-overlapping subgroups that are called clusters.
- K-means is a centroid-based algorithm or a distance-based algorithm, where we calculate the distances to assign a point to a cluster.
- In K-Means, each cluster is associated with a centroid.

K-Means Clustering

1. **Initialization**: Begin by randomly selecting K data points from the dataset. These selected points serve as the initial centroids of the clusters.
2. **Assignment**: For each data point, compute the distance between the point and each of the K centroids. Assign the data point to the cluster with the nearest centroid. This process results in the formation of K clusters.
3. **Update centroids**: After assigning all data points, recompute the centroid of each cluster by calculating the mean of the data points within that cluster.
4. **Iteration:** Repeat steps 2 and 3 until the algorithm converges. Convergence occurs when the centroids no longer change significantly or when a specified number of iterations is reached.
5. **Final Output**: Once convergence occurs, the algorithm produces the final cluster centroids along with the cluster membership of each data point.

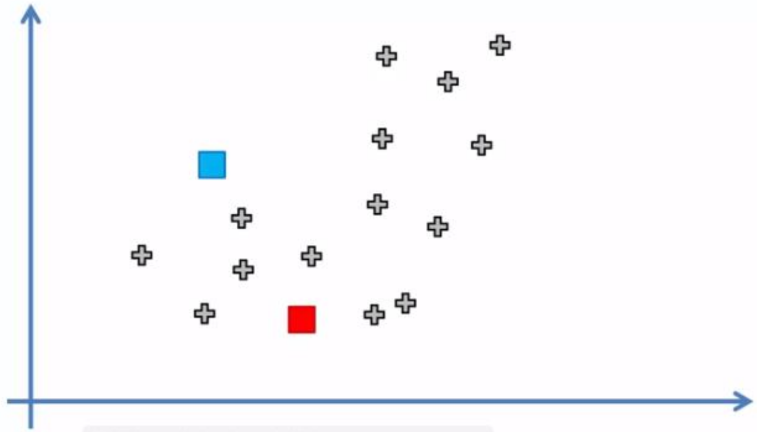
K-Means Clustering

STEP 1: Choose the number K of clusters: K = 2



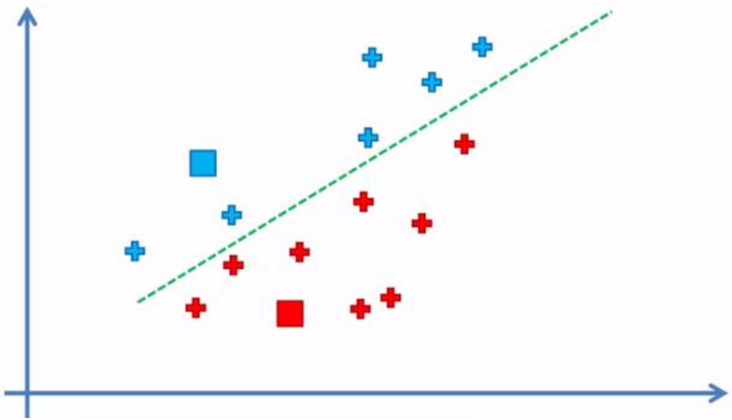
K-Means Clustering

STEP 2: Select at random K points, the centroids (not necessarily from your dataset)



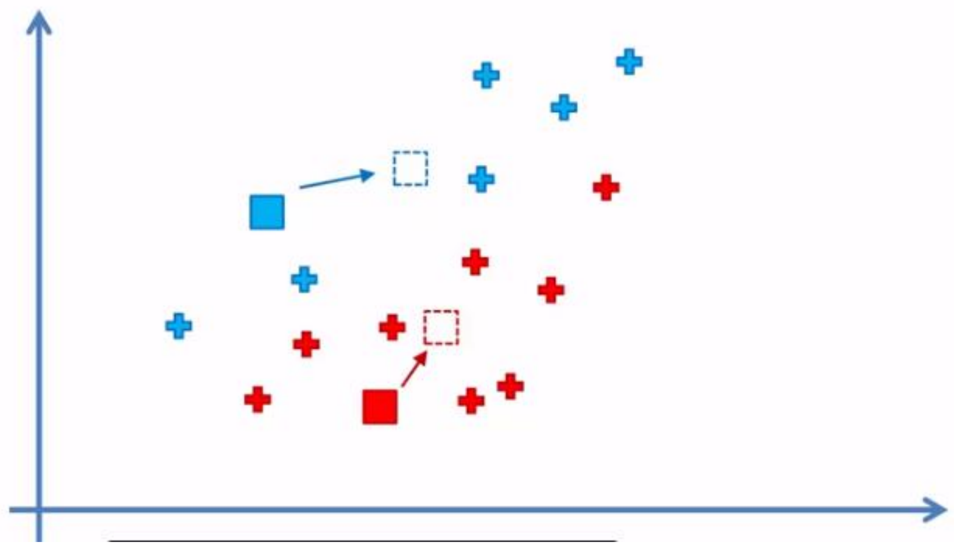
K-Means Clustering

STEP 3: Assign each data point to the closest centroid ➡ That forms K clusters



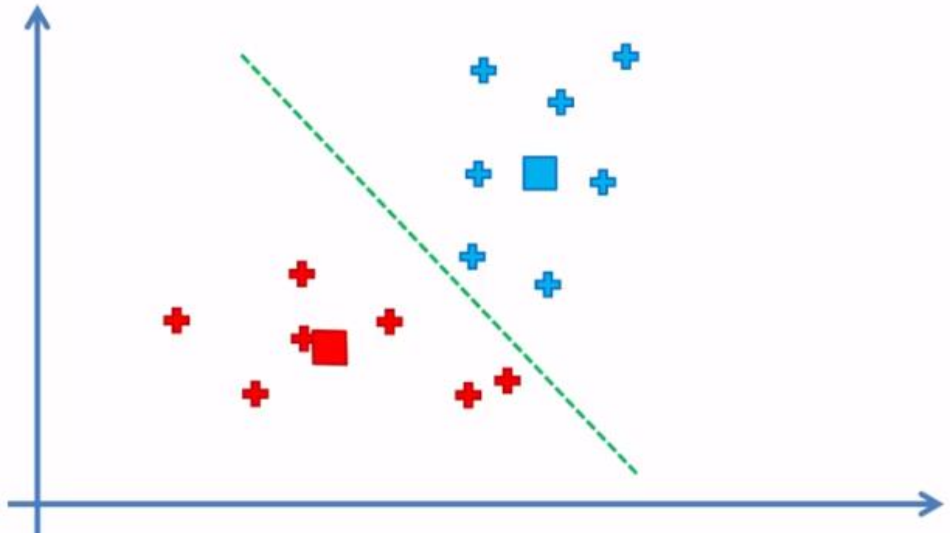
K-Means Clustering

STEP 4: Compute and place the new centroid of each cluster



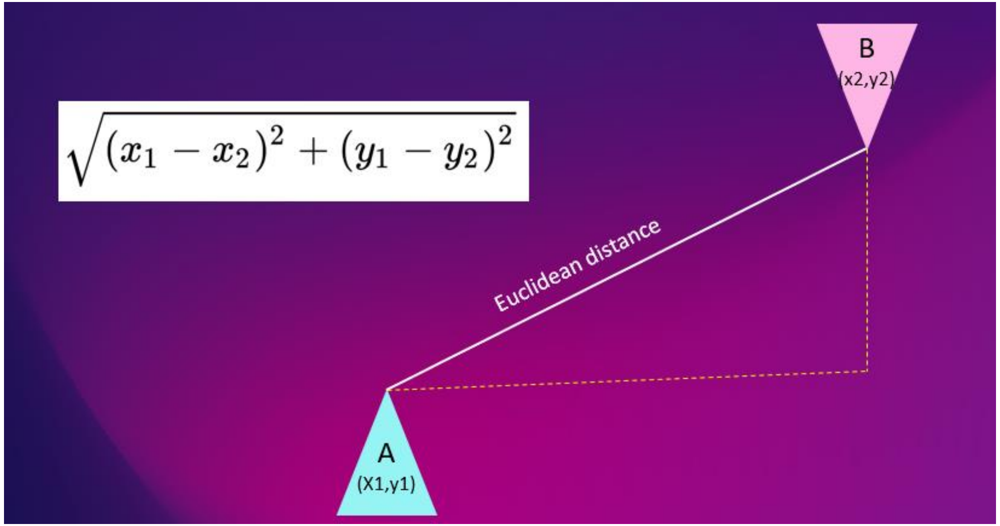
K-Means Clustering

STEP 5: Reassign each data point to the new closest centroid.
If any reassignment took place, go to STEP 4, otherwise go to FIN.



K-Means Clustering

- Distance measure

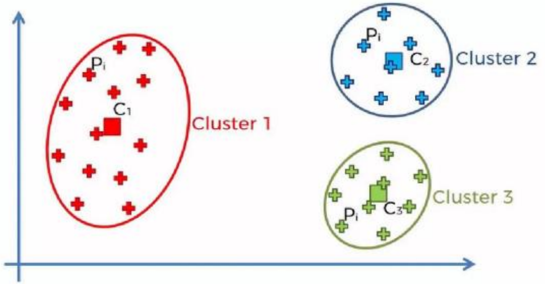


Euclidean Distance = √((X1 — X2)² + (Y1 — Y2)²)

K-Means Clustering

Choosing the right K

- The way to evaluate the choice of K is made using a parameter known as **WCSS** (Within Cluster Sum of Squares).
- It should be low. Here's the formula representation for example when K = 3



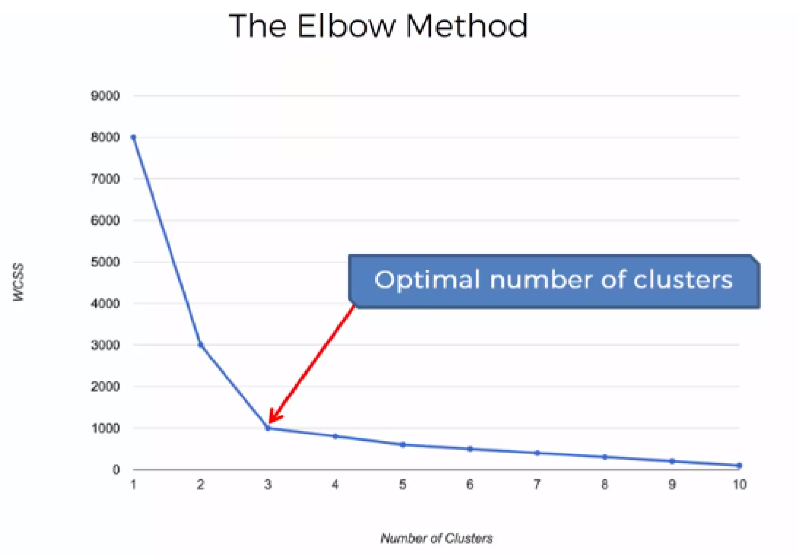
$$\mathrm{WCSS} = \sum_{P_i \text{ in Cluster } 1} \mathrm{distance}(P_i, C_1)^2 + \sum_{P_i \text{ in Cluster } 2} \mathrm{distance}(P_i, C_2)^2 + \sum_{P_i \text{ in Cluster } 3} \mathrm{distance}(P_i, C_3)^2$$

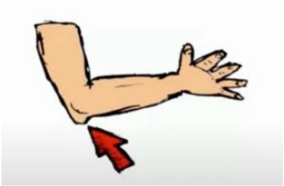
Summation Distance(p,c) is the sum of distance of points in a cluster from the centroid.

K-Means Clustering

Elbow Method

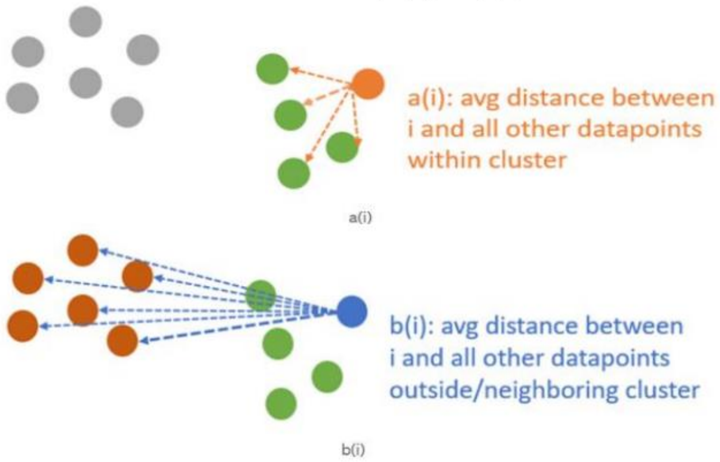
- The Elbow method is then used to choose the best K value.





Evaluation metric : Silhouette score

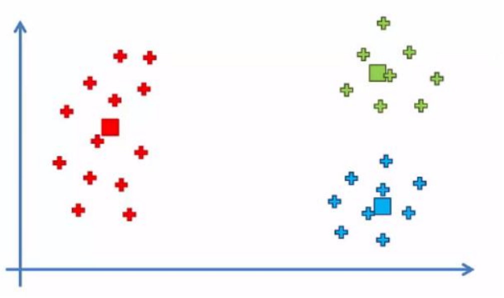
$$s(i) = \frac{b(i) - a(i)}{\max\left(a(i), b(i)\right)}$$



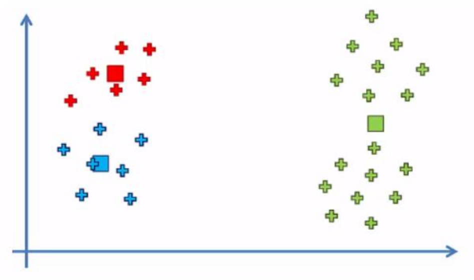
- The value of the silhouette score (S(i)) is between [-1, 1].
- A score of 1 denotes the best.
- The worst value is -1.
- Values near 0 denote overlapping clusters.

Centroid Random Initialization

Init 1



Init 2

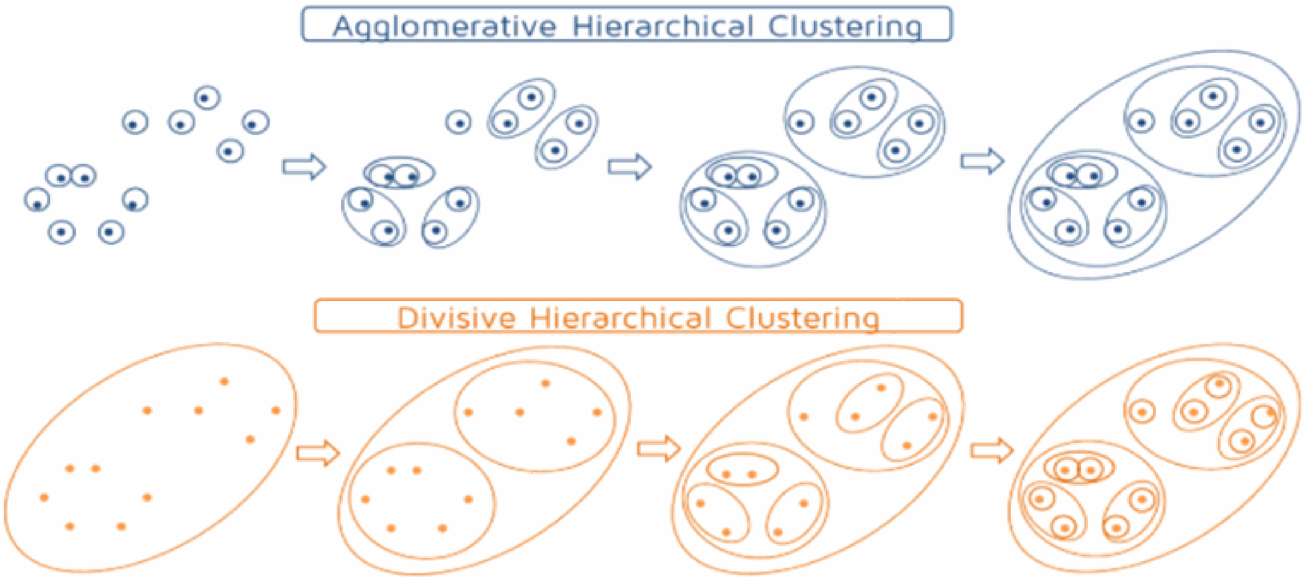


KMeans++.

Hierarchical Clustering

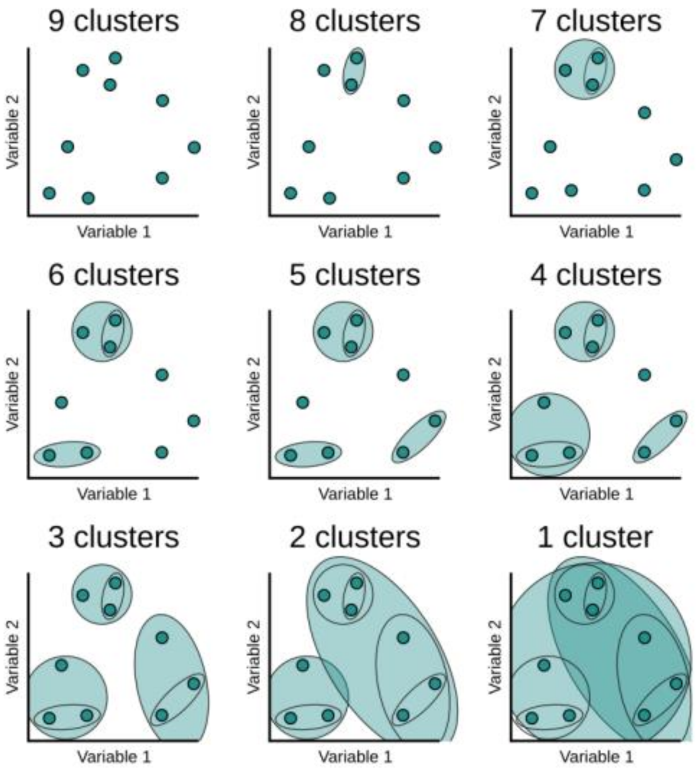
- It creates a hierarchy of clusters by merging or splitting them based on similarity measures.
- It uses a bottom-up or top-down approach to construct a hierarchical data clustering schema.
- There are two common hierarchical clustering algorithms, each with its own approach:
    - Agglomerative Clustering
    - Divisive Clustering

Hierarchical Clustering



Agglomerative Clustering

- Agglomerative hierarchical clustering merges clusters that are closest to each other at each iteration.
- As the algorithm progresses, pairs of the closest clusters are successively merged, forming a dendrogram — a tree-like structure representing the hierarchical relationships between clusters.
- Ellipses indicate the formation of clusters at each iteration, going from top left to bottom right



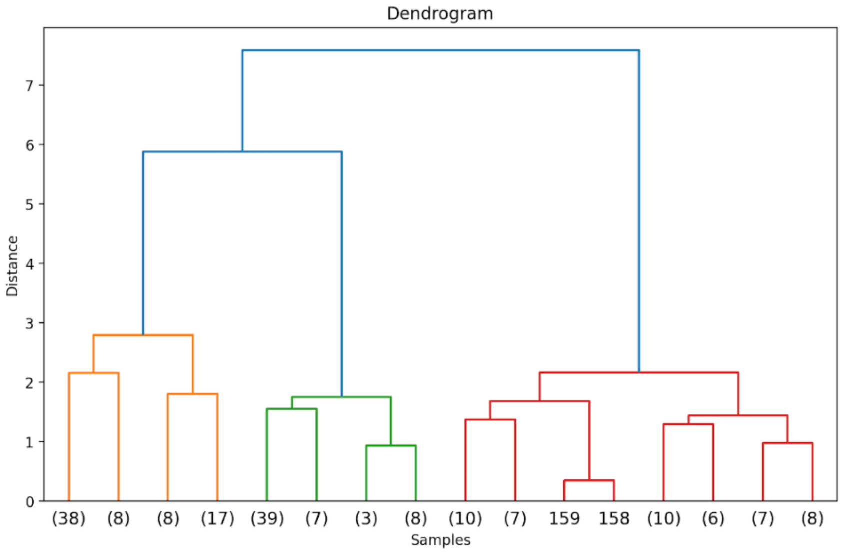
Agglomerative Clustering

Suppose there are n distinct data points in the dataset. Agglomerative clustering works as follows:

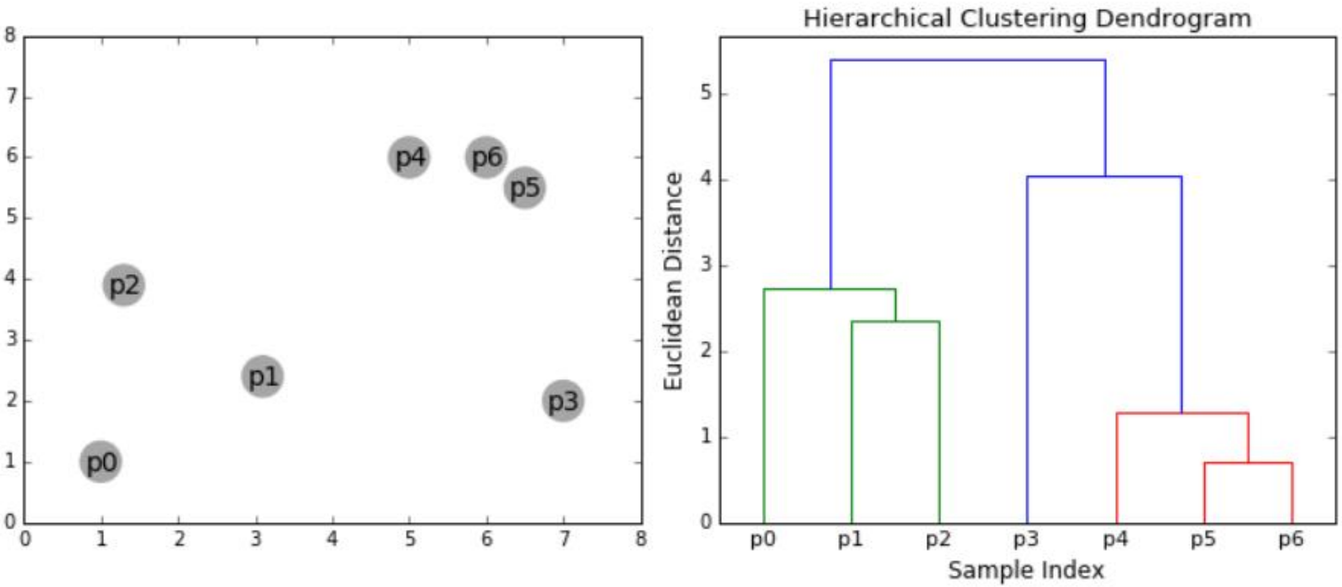
1. Start with n clusters; each data point is a cluster in itself.
2. Group data points together based on similarity between them.
3. Meaning similar clusters are merged depending on the distance.
4. Repeat step 2 until there is only one cluster.

Dendrogram

- It is a hierarchical tree structure that helps us understand how the data points and subsequently clusters are grouped or merged together as the algorithm proceeds.
- We can visualize the result of clustering as a dendrogram.
- In the hierarchical tree structure, the leaves denote the instances or the data points in the data set.
- The corresponding distances at which the merging or grouping occurs can be inferred from the y-axis.



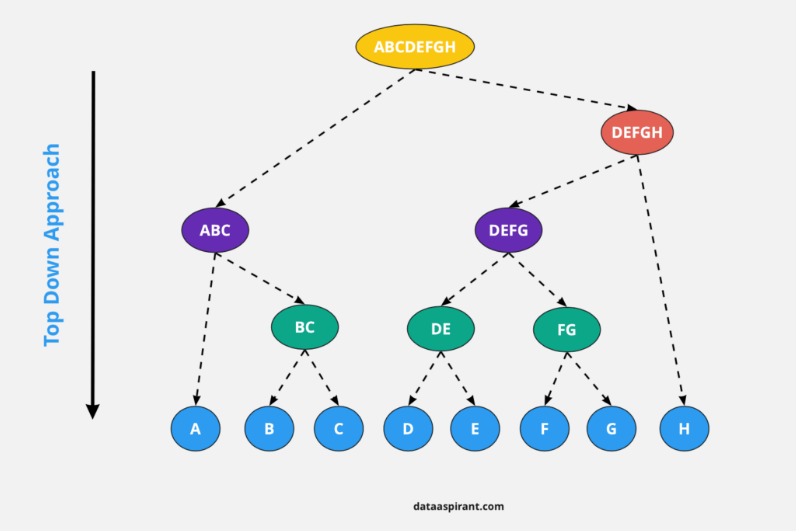
Agglomerative Clustering



Divisive Clustering

As the name suggests, divisive clustering tries to perform the inverse of agglomerative clustering:

1. All the n data points are in a single cluster.
2. Divide this single large cluster into smaller groups. Note that the grouping together of data points in agglomerative clustering is based on similarity. But splitting them into different clusters is based on dissimilarity; data points in different clusters are dissimilar to each other.
3. Repeat until each data point is a cluster in itself.

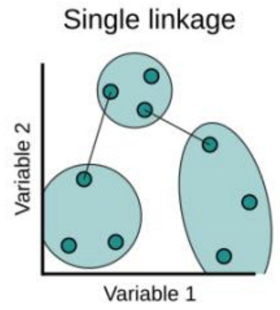


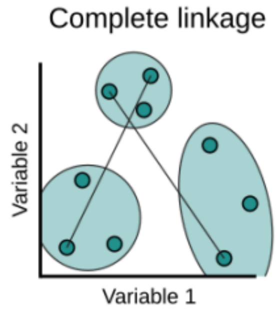
Distance Between Clusters : Cluster Linkage

- Determines how clusters are merged (agglomerative) or split (divisive) based on their dissimilarity.
- An appropriate distance metric (e.g., Euclidean distance) to measure similarity between individual observations is required
- Specifies how distances between clusters are calculated using a linkage criterion.
- Commonly used linkage criteria include:
  - Single linkage
  - Complete linkage
  - Average linkage
  - Ward’s linkage

Linkage Types

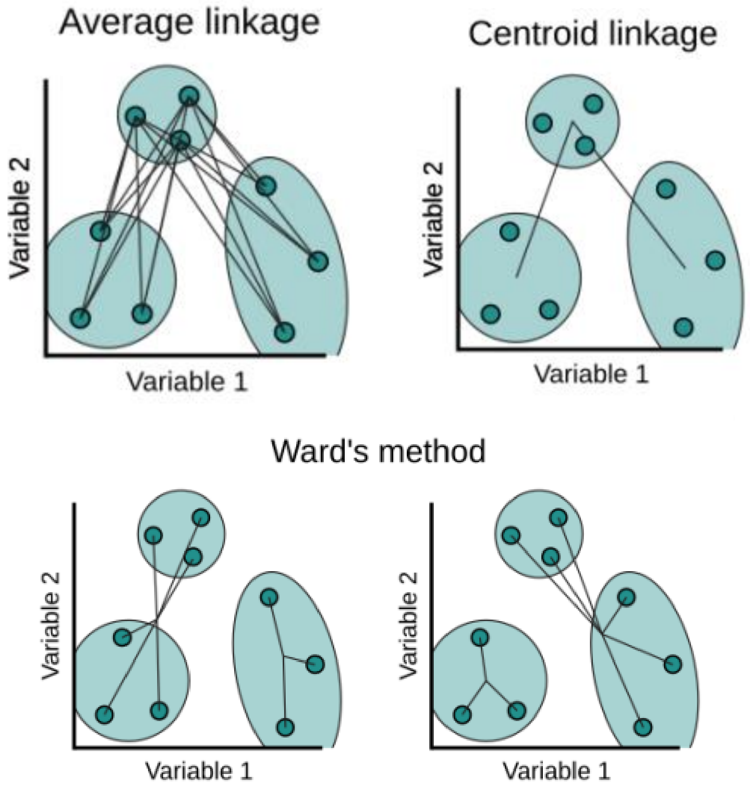
- **Single (Min) linkage -**Computes the shortest distance between any two points in two clusters. The two clusters with the smallest minimum distance are merged.
- **Complete (Max) linkage -** Computes the maximum distance between any two points in different clusters. The two clusters where this maximum distance is smallest compared to all other possible merges are merged.





Linkage Types

- **Average linkage -** Computes the average distance between all pairs of points in two clusters. The two clusters with the smallest average distance are merged.
- **Centroid Linkage -** Computes the distance between the centroids of two clusters. The two clusters with the smallest distance between their centroids are merged.
- **Ward's linkage -** Merges the two clusters that minimize the increase in the total within cluster variance



Key factors for selecting the linkage method

- Single Linkage: Can handle non-convex shaped clusters and is sensitive to outliers and noise, but it can also create clusters that are not well-separated.
- Complete Linkage: Tends to form compact, well-separated clusters, but it can be less sensitive to outliers and can create long chains of clusters. However, MAX can break large clusters and tends to be biased towards globular clusters.
- Average Linkage: Compromise between complete and single linkage and often produces balanced clusters.
- Ward's Method: It tends to create clusters with similar sizes and compact shapes, making it suitable for cases where the goal is to minimize the variance within clusters.

Key factors for selecting the linkage method

- **Understand your data**: Distribution of data points, the presence of outliers, and the nature of the clusters you expect to find. Certain methods may perform better with different data structures.
- **Visualize the dendrogram**: Look for compact, well-separated clusters that make sense for your data.
- **Evaluate cluster validity**: Silhouette score and within-cluster sum of squares (WCSS).
- **Consider computational efficiency**: Some proximity methods may be computationally more expensive than other

References

- https://www.kaggle.com/code/shrutimechlearn/step-by-step-kmeans-explained-in-detail
- https://www.kdnuggets.com/unveiling-hidden-patterns-an-introduction-to-hierarchical-clustering